

| ConsoleBookstore                     |
|--------------------------------------|
| +MemberService ms                    |
| +BookService bs                      |
| +OrderService os                     |
| +DeliveryService ds                  |
| +CartService cs                      |
| +start_menu                          |
| +member_menu                         |
| +myinfo_menu                         |
| +book_browse_menu                    |
| +cart_view_menu                      |
| +admin_menu                          |
| +book_admin_menu                     |
| +order_admin_menu                    |
| +delivery_admin_menu                 |
| +member_admin_menu                   |
| +main()                              |
| +show_welcome()                      |
| +say_goodbye()                       |
| +print_title(title)                  |
| +input_int(prompt)                   |
| +select_menu(menu_list)              |
| +run_start_menu()                    |
| +menu_login()                        |
| +menu_join()                         |
| +menu_list_books()                   |
| +menu_book_info()                    |
| +menu_logout()                       |
| +run_member_menu()                   |
| +run_book_browse_menu()              |
| +menu_add_to_cart()                  |
| +menu_view_order()                   |
| +menu_list_orders()                  |
| +menu_cancel_order()                 |
| - __cancel_linked_delivery(order_id) |
| +menu_view_delivery()                |
| +menu_quit_membership()              |
| +run_cart_view_menu()                |
| +menu_order_cart()                   |
| +menu_delete_cart()                  |
| +menu_view_cart()                    |
| +run_myinfo_menu()                   |
| +menu_view_myinfo()                  |
| +menu_update_password()              |
| +menu_update_member_info()           |
| +run_admin_menu()                    |
| +run_book_admin_menu()               |
| +menu_add_book()                     |
| +menu_update_book()                  |
| +menu_delete_book()                  |
| +run_order_admin_menu()              |
| +menu_list_orders_admin()            |
| +menu_view_order_admin()             |
| +menu_cancel_order_admin()           |
| +run_delivery_admin_menu()           |
| +menu_view_delivery_admin()          |
| +menu_update_delivery_status()       |
| +run_member_admin_menu()             |
| +menu_list_members()                 |
| +menu_quit_member_admin()            |

| MemberService          |
|------------------------|
| +String ADMIN_ID       |
| +String ADMIN_PASSWORD |
| -Object dao            |
| +Object current_user   |
| +join()                |
| +login()               |
| +logout()              |
| +is_admin()            |
| +list_members()        |
| +view_member_info()    |
| +update_member_info()  |
| +update_password()     |
| +remove_member()       |
| Member                 |
| -int id                |
| -string password       |
| -string name           |
| -string address        |
| -string phone          |
| +getters()             |
| +setters()             |
| MemberDAO              |
| -memberDB              |
| +insert_member()       |
| +select_member_by_id() |
| +select_all_members()  |
| +update_member()       |
| +delete_member()       |

| OrderService                                    |
|-------------------------------------------------|
| -Object dao                                     |
| -Object book_dao                                |
| +create_order()                                 |
| +get_order()                                    |
| +list_orders()                                  |
| +list_orders_by_member()                        |
| +cancel_order(order_id, requester_id, is_admin) |
| Order                                           |
| -int order_id                                   |
| -int member_id                                  |
| -List items                                     |
| -datetime order_date                            |
| -string status                                  |
| +getters()                                      |
| +setters()                                      |
| + __str__()                                     |
| OrderDAO                                        |
| -orderDB                                        |
| +insert_order()                                 |
| +select_order_by_id()                           |
| +select_all_orders()                            |
| +select_orders_by_member()                      |
| +update_order()                                 |
| +delete_order()                                 |
| OrderItem                                       |
| -int order_id                                   |
| -int book_id                                    |
| -int quantity                                   |
| +getters()                                      |
| + __str__()                                     |
| OrderItemDAO                                    |
| -orderDB                                        |
| +insert_order_item(order_item)                  |
| +select_order_items_by_order_id(order_id)       |
| +select_all_order_items()                       |
| +update_order_item(order_item_id, order_item)   |
| +delete_order_item(order_item_id)               |

| BookService          |
|----------------------|
| -Object dao          |
| +add_book()          |
| +update_book()       |
| +remove_book()       |
| +get_book()          |
| +list_books()        |
| Book                 |
| -int book_id         |
| -string title        |
| -string author       |
| -float price         |
| -int stock           |
| +getters()           |
| +setters()           |
| + __str__()          |
| BookDAO              |
| -bookDB              |
| +insert_book()       |
| +select_book_by_id() |
| +select_all_books()  |
| +update_book()       |
| +delete_book()       |

| DeliveryService                                |
|------------------------------------------------|
| -Object dao                                    |
| +create_delivery(order_id, member_id, address) |
| +get_delivery()                                |
| +get_delivery_by_order()                       |
| +get_delivery_by_member()                      |
| +list_deliveries()                             |
| +update_status()                               |
| Delivery                                       |
| -int delivery_id                               |
| -int order_id                                  |
| -int member_id                                 |
| -string address                                |
| -string status                                 |
| -datetime delivery_date                        |
| +getters()                                     |
| +set_status()                                  |
| +set_date()                                    |
| + __str__()                                    |
| DeliveryDAO                                    |
| -deliveryDB                                    |
| +insert_delivery()                             |
| +select_delivery_by_id()                       |
| +select_by_order()                             |
| +select_by_member()                            |
| +select_all_deliveries()                       |
| +update_delivery()                             |
| + __str__()                                    |

| CartService                                     |
|-------------------------------------------------|
| -Object dao                                     |
| +add_to_cart()                                  |
| +add_item_to_cart(member_id, book_id, quantity) |
| +view_cart()                                    |
| +delete_cart()                                  |
| Cart                                            |
| -int cart_id                                    |
| +select_all_carts()                             |
| +setters()                                      |
| CartDAO                                         |
| -cartDB                                         |
| +insert_cart()                                  |
| +select_cart_by_member()                        |
| +select_all_carts()                             |
| +delete_cart()                                  |
| CartItem                                        |
| -int cart_id                                    |
| -int member_id                                  |
| -int book_id                                    |
| -int quantity                                   |
| +getters()                                      |
| +setters()                                      |
| + __str__()                                     |
| CartItemDAO                                     |
| -cartDB                                         |
| +insert_cart(cart)                              |
| +select_cart_by_member(member_id)               |
| +select_all_carts()                             |
| +delete_cart(cart_id)                           |